

Answer Key: Practice Test 1 (The Special Senses)

Multiple Choice

1. **c) Pain**
 2. **b) Chemoreceptor**
 3. **b) Only cones**
 4. **b) CN VII (Facial)**
 5. **c) Semicircular canals**
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True/False

6. **False** – The olfactory epithelium is located in the superior part of the nasal cavity.
 7. **False** – Bitter tastes are detected by G-protein-coupled receptors, not ion channels.
 8. **True**
 9. **True**
 10. **False** – The macula lutea has the highest concentration of cones, not rods.
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Fill in the Blank

11. **Bowman's**
 12. **Vestibulocochlear (CN VIII)**
 13. **Cornea**
 14. **Static, dynamic**
 15. **Spiral organ (Organ of Corti)**
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Short Answer

16. **Cyclic AMP (cAMP) in Olfaction:** Odorant molecules bind to receptors on olfactory hairs, activating G-proteins. This leads to the production of cAMP, which opens ion channels, causing depolarization and triggering nerve impulses.
17. **Eustachian Tube Function:** It equalizes pressure between the middle ear and the external environment, preventing damage to the tympanic membrane during pressure changes.
18. **Five Primary Tastes:**
 - Sour: H⁺ ions (acids).
 - Sweet: Sugars and artificial sweeteners.
 - Bitter: Alkaloids (e.g., caffeine).

- Salty: Na⁺ ions.
 - Umami: Glutamate (savory/meaty flavor).
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Answer Key: Practice Test 2 (Endocrine System)

Multiple Choice

1. **c) Cortisol**
 2. **b) Releasing and inhibiting hormones**
 3. **a) Oxytocin**
 4. **b) Decrease blood calcium levels**
 5. **b) Epinephrine and norepinephrine**
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True/False

6. **False** – Water-soluble hormones bind to cell membrane receptors, not intracellular receptors.
 7. **False** – The posterior pituitary stores and releases oxytocin and ADH but does not produce them (they are produced in the hypothalamus).
 8. **True**
 9. **False** – Insulin decreases blood glucose levels.
 10. **True**
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Fill in the Blank

11. **Transport**
 12. **Hypophyseal**
 13. **Cortisol**
 14. **Pineal**
 15. **Antidiuretic (ADH)**
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Short Answer

16. **Lipid-Soluble vs. Water-Soluble Hormones:**
 - Lipid-soluble hormones diffuse through the plasma membrane and bind to intracellular receptors, altering gene expression.
 - Water-soluble hormones bind to cell surface receptors, activating second messengers like cAMP to trigger cellular responses.

17. Three Zones of Adrenal Cortex:

- Zona glomerulosa: Secretes mineralocorticoids (e.g., aldosterone).
- Zona fasciculata: Secretes glucocorticoids (e.g., cortisol).
- Zona reticularis: Secretes androgens (e.g., DHEA).

18. Thyroid Hormone Regulation:

- Low T3/T4 levels stimulate the hypothalamus to release TRH → TRH stimulates the anterior pituitary to release TSH → TSH stimulates the thyroid gland to secrete T3/T4. High T3/T4 levels inhibit TRH and TSH secretion (negative feedback).

Answer Key: Practice Test 3 (Reproductive System)

Multiple Choice

1. **b) Form the blood-testis barrier**
2. **b) Metaphase II**
3. **b) LH**
4. **b) Progesterone**
5. **c) Fallopian tube**

True/False

6. **False** – Spermatogenesis occurs in the seminiferous tubules of the testes.
7. **True**
8. **False** – The myometrium is the middle layer of the uterus; the endometrium is the innermost layer.
9. **False** – The stratum functionalis is shed during menstruation, not the stratum basalis.
10. **True**

Fill in the Blank

11. **LH (Luteinizing Hormone)**
12. **Infundibulum**
13. **Ovarian**
14. **Proliferative**
15. **Prostate**

Short Answer

16. Pathway of Sperm:

- Seminiferous tubules → straight tubules → rete testis → efferent ducts → epididymis → vas deferens → ejaculatory duct → urethra → ejaculation.

17. Four Phases of Female Reproductive Cycle:

- Menstrual Phase (Days 1–5): Shedding of the stratum functionalis.
- Preovulatory Phase (Days 6–13): Follicular growth and endometrial repair.
- Ovulation (Day 14): Release of secondary oocyte due to LH surge.
- Postovulatory Phase (Days 15–28): Corpus luteum secretes progesterone and estrogen; if no fertilization, hormone levels drop, leading to menstruation.

18. Hormonal Changes Without Fertilization:

- If fertilization does not occur, the corpus luteum degenerates into the corpus albicans. Progesterone and estrogen levels drop, triggering the shedding of the endometrium (menstruation). FSH and LH levels begin to rise again to start a new cycle.